Evaluation of Large Language Models: Questions and Answers

Based on EECS 836 Machine Learning Notes

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Introduction

This document provides a comprehensive set of fundamental, computational, and conceptual questions and answers on the evaluation of large language models (LLMs), covering topics such as benchmark datasets (e.g., MMLU, BIG-bench, MT-Bench), challenges in multiple-choice and open-ended question evaluation, human evaluation platforms (e.g., Chatbot Arena), and ethical considerations (e.g., MACHIAVELLI benchmark). The content is derived from the lecture notes of EECS 836: Machine Learning by Zijun Yao at the University of Kansas, with slides adopted from Hongyi Lee@NTU.

1 Fundamental Questions

These questions focus on the basic concepts and definitions related to LLM evaluation.

1. **What is the Massive Multitask Language Understanding (MMLU) benchmark?**
   - *Answer*: The MMLU is a benchmark designed to evaluate a language model's ability to perform well across a wide range of subjects, including STEM, humanities, and professional fields. It consists of multiple-choice questions, such as those in mathematics, physics, and chemistry, to test the model's knowledge and reasoning capabilities (e.g., solving polynomial ring problems in $\mathbb{Z}_3$).

2. **What is the BIG-bench benchmark, and what types of tasks does it include?**
   - *Answer*: BIG-bench (Beyond the Imitation Game Benchmark) is a comprehensive benchmark for evaluating LLMs on diverse tasks. It includes creative tasks like identifying movies from emojis, strategic tasks like finding checkmate-in-one chess moves, and pattern recognition tasks like decoding ASCII art (e.g., recognizing the word "BENCH" from ASCII).

3. **What is the purpose of the Chatbot Arena?**
   - *Answer*: Chatbot Arena, hosted at `https://lmarena.ai/`, is a platform for human evaluation of LLMs. It allows users to interact with multiple models

simultaneously, compare their responses to open-ended questions (e.g., how to increase work efficiency), and vote on which model provides better answers, generating a leaderboard based on human preferences.

4. **What is the Needle in a Haystack test for LLMs?**
   - *Answer*: The Needle in a Haystack test evaluates an LLM's ability to retrieve specific facts from long contexts. For example, it tests whether a model can identify the most fun activity in San Francisco when the relevant information is buried deep in a lengthy document, assessing context length and document depth handling (e.g., GPT-4 128K, Claude-2.1 200K).

5. **What is the MACHIAVELLI benchmark, and what does it measure?**
   - *Answer*: The MACHIAVELLI benchmark assesses the ethical behavior of LLMs in interactive, game-like scenarios. It measures metrics like deception, stealing, physical harm, and utility to others, evaluating whether models prioritize ethical decisions or pursue goals at the expense of ethical violations (e.g., choosing to deceive in a scenario to advance ambitions).

2 Computational Questions

These questions involve calculations, data analysis, or algorithmic procedures related to LLM evaluation.

1. **Given the MMLU scores for llama-65b across different implementations (HELM: 0.637, Harness: 0.488, Harness: 0.636), calculate the average score.**
   - *Answer*: To find the average MMLU score for llama-65b:

   $$\text{Average} = \frac{0.637 + 0.488 + 0.636}{3} = \frac{1.761}{3} \approx 0.587$$

   Thus, the average MMLU score is approximately 0.587.

2. **In the MMLU question "Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2+c)$ is a field," the correct answer is B (1). If a model outputs "3" (D), is it correct or wrong?**
   - *Answer*: The correct answer is B (1), as $\mathbb{Z}_3[x]/(x^2+1)$ is a field (the polynomial x^2+1 is irreducible over $\mathbb{Z}_3$). If the model outputs "3" (D), it is *wrong*, as $x^2+3 \equiv x^2$ (since $3 \equiv 0 \mod 3$), and $\mathbb{Z}_3[x]/(x^2)$ is not a field (it has zero divisors).

3. **Using the Chatbot Arena data for vicuna-v1.3-33B (Original: 57.0, A: 59.5, B: 58.6, C: 64.8), calculate the performance drop when moving the correct answer to C.**

- *Answer*: The original performance is 57.0, and when the correct answer is moved to C, the performance is 64.8. The performance change is:

$$\text{Change} = 64.8 - 57.0 = 7.8$$

This represents a performance *increase* of 7.8 points, not a drop.

4. **In the Needle in a Haystack test for Claude-2.1 200K, if the accuracy of fact retrieval is 50% at 50% document depth, what is the expected number of correct retrievals out of 100 attempts?**

    - *Answer*: The accuracy is 50%, so the expected number of correct retrievals out of 100 attempts is:

    $$\text{Expected correct} = 100 \times 0.5 = 50$$

    Thus, 50 correct retrievals are expected.

5. **In the MACHIAVELLI benchmark, a model scores 4 on deception and 71 on physical harm. If these are weighted equally in an ethical violation score, compute the total ethical violation score.**

    - *Answer*: Assuming equal weights for deception and physical harm:

    $$\text{Total ethical violation score} = 4 + 71 = 75$$

    Thus, the total ethical violation score is 75.

3 Conceptual Questions

These questions explore the deeper understanding and implications of LLM evaluation techniques.

1. **Why do different implementations of the MMLU benchmark yield varying scores for the same model (e.g., llama-65b: 0.637 vs. 0.488)?**

    - *Answer*: Variations in MMLU scores across implementations (e.g., HELM vs. Harness) arise due to differences in evaluation protocols, such as prompt formatting, answer parsing, or subset of questions used. For example, some implementations may misinterpret model outputs (e.g., "b" vs. "1" for the same choice) or handle edge cases differently, highlighting the challenge of standardizing multiple-choice question evaluation.

2. **Why might moving the correct answer to option A improve model performance in multiple-choice benchmarks?**

    - *Answer*: Moving the correct answer to option A can improve performance due to positional bias in LLMs, where models may favor earlier options (e.g., A) due to training data patterns or attention mechanisms. The document shows models like vicuna-v1.3-33B perform better when correct answers are in A (59.5 vs. 57.0), suggesting that evaluation design can influence results.

3. **How does human evaluation in Chatbot Arena differ from automated benchmark evaluation like MMLU?**

    - *Answer*: Human evaluation in Chatbot Arena relies on subjective user preferences, comparing model responses to open-ended questions (e.g., work efficiency tips) and ranking them based on quality, relevance, or clarity. In contrast, MMLU uses objective, multiple-choice questions with definitive answers, scored automatically. Human evaluation captures nuanced aspects like coherence but is subjective, while automated benchmarks are consistent but limited to predefined tasks.

4. **What are the challenges of evaluating LLMs on open-ended tasks like translation or summarization?**

    - *Answer*: Open-ended tasks lack a single standard answer, making evaluation subjective. For example, translations can vary in phrasing but still be correct, and summaries may differ in focus but remain valid. Automated metrics (e.g., BLEU for translation) often fail to capture semantic equivalence, while human evaluation is costly and inconsistent, as shown in the document's discussion of tasks without standard answers.

5. **Why is the MACHIAVELLI benchmark important for evaluating LLMs?**

    - *Answer*: The MACHIAVELLI benchmark is crucial because it evaluates LLMs beyond task performance, focusing on ethical behavior in decision-making scenarios. By measuring deception, harm, and utility, it reveals whether models prioritize goals unethically (e.g., deceiving to advance ambitions), which is vital for ensuring safe and trustworthy AI deployment in real-world applications.

6. **What is the significance of the Theory of Mind evaluation for LLMs?**

7. **Why should benchmark results be treated with caution, as highlighted in the document?**

    - *Answer*: Benchmark results may be unreliable due to data contamination, where models are trained on datasets containing benchmark questions (e.g., MMLU questions in GPT-3.5-T training data). Additionally, rephrased questions can lead to different scores (e.g., Llama-2-13B on MMLU vs. rephrased MMLU), and evaluation inconsistencies (e.g., MMLU implementation variations) further erode trust. This suggests that benchmark scores may not fully reflect a model's true capabilities.

8. **How does the Needle in a Haystack test reveal limitations in LLMs' context handling?**

    - *Answer*: The Needle in a Haystack test shows that LLMs struggle to retrieve facts from long contexts, especially at greater document depths (e.g., Claude-2.1's 50% accuracy at 50